

ANALYTIC ASPECTS OF QUADRATIC FORMS (CTNT 2026)

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1. INTRODUCTION

A *quadratic form* is a polynomial that is homogeneous of degree 2, such as

$$x^2 + y^2, \quad 5x^2 - 14xy + 10y^2, \quad \frac{7}{3}xy + \frac{15}{8}xz - \frac{12}{5}yz, \quad x_1^2 + x_2^2 + \cdots + x_k^2.$$

The first two examples above depend on two variables, so they are called binary. The third example depends on three variables and is called ternary. A quadratic form in k variables, called a k -dimensional quadratic form, will be written as $Q(x_1, \dots, x_k)$ or $Q(\mathbf{x})$ when $\mathbf{x}$ is understood to mean $(x_1, \dots, x_k)$. We will focus on quadratic forms with coefficients in $\mathbf{Z}$, which are called *integral* quadratic forms. Similarly, *rational* quadratic forms have coefficients in $\mathbf{Q}$.

A lot of classical number theory is concerned with the behavior of integral quadratic forms. Here are three such results:

- describe primes of the form $x^2 + y^2$ where $x, y \in \mathbf{Z}$ (Fermat's two-square theorem),
- describe the integral solutions of $x^2 - dy^2 = 1$ for $d \in \mathbf{Z}$ that is not a perfect square (Pell's equation),
- show every nonnegative integer is $a^2 + b^2 + c^2 + d^2$ for some $a, b, c, d \in \mathbf{Z}$ (Lagrange's four-square theorem).¹

The simplest k -dimensional quadratic form is $x_1^2 + x_2^2 + \cdots + x_k^2$, which is the dot product $\mathbf{x} \cdot \mathbf{x}$. When a k -dimensional quadratic form has coefficients in a field *not* of characteristic 2, we can write it in terms of the dot product and a $k \times k$ symmetric matrix, as follows.

Theorem 1.1. *Over a field of characteristic not equal to 2, a k -dimensional quadratic form $Q(\mathbf{x})$ is $\mathbf{x} \cdot A\mathbf{x}$ for a unique $k \times k$ symmetric matrix A .*

Proof. We have

$$\begin{aligned} Q(x_1, \dots, x_k) &= \sum_{i=1}^k a_i x_i^2 + \sum_{1 \leq i < j \leq k} a_{ij} x_i x_j \\ &= \sum_{i=1}^k a_i x_i^2 + \sum_{1 \leq i < j \leq k} \left(\frac{a_{ij}}{2} x_i x_j + \frac{a_{ij}}{2} x_j x_i \right) \\ (1.1) \quad &= \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} \cdot \begin{pmatrix} a_1 & a_{12}/2 & \cdots & a_{1k}/2 \\ a_{12}/2 & a_2 & \cdots & a_{2k}/2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{1k}/2 & a_{2k}/2 & \cdots & a_k \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix}, \end{aligned}$$

¹Section 6 of <https://kconrad.math.uconn.edu/blurbs/ugradnumthy/descent.pdf> has a proof of the four-square theorem by descent.

which is $\mathbf{x} \cdot A\mathbf{x}$ where A is the symmetric matrix in (1.1).

Why is there just one $k \times k$ symmetric matrix A such that $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ for all $\mathbf{x}$? Suppose $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ and $Q(\mathbf{x}) = \mathbf{x} \cdot A'\mathbf{x}$ for all $\mathbf{x}$, where A and A' are both $k \times k$ symmetric matrices, so $\mathbf{x} \cdot A\mathbf{x} = \mathbf{x} \cdot A'\mathbf{x}$ for all $\mathbf{x}$. We want to show $A = A'$ and will use the trick of replacing $\mathbf{x}$ with $\mathbf{x} + \mathbf{y}$: $(\mathbf{x} + \mathbf{y}) \cdot A(\mathbf{x} + \mathbf{y}) = (\mathbf{x} + \mathbf{y}) \cdot A'(\mathbf{x} + \mathbf{y})$ for all $\mathbf{x}$ and $\mathbf{y}$, and expanding both sides gives us

$$\mathbf{x} \cdot A\mathbf{x} + \mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} + \mathbf{y} \cdot A\mathbf{y} = \mathbf{x} \cdot A'\mathbf{x} + \mathbf{x} \cdot A'\mathbf{y} + \mathbf{y} \cdot A'\mathbf{x} + \mathbf{y} \cdot A'\mathbf{y}.$$

Canceling the equal first terms and the equal last terms on both sides,

$$(1.2) \quad \mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} = \mathbf{x} \cdot A'\mathbf{y} + \mathbf{y} \cdot A'\mathbf{x}$$

A *fundamental* property links the dot product and matrix transpose: for $k \times k$ matrices M , $\boxed{M\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot M^\top \mathbf{y}}$. Thus when M is symmetric, so $M^\top = M$, we get $M\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot M\mathbf{y}$: a symmetric matrix can be moved to the other side of a dot product. Apply this to (1.2): since $A^\top = A$,

$$\mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} = \mathbf{x} \cdot A\mathbf{y} + A\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot A\mathbf{y} + \mathbf{x} \cdot A\mathbf{y} = 2\mathbf{x} \cdot A\mathbf{y}.$$

Similarly, $\mathbf{x} \cdot A'\mathbf{y} + \mathbf{y} \cdot A'\mathbf{x} = 2\mathbf{x} \cdot A'\mathbf{y}$, so $2\mathbf{x} \cdot A\mathbf{y} = 2\mathbf{x} \cdot A'\mathbf{y}$ and thus $\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot A'\mathbf{y}$. This equation holds for all $\mathbf{x}$ and $\mathbf{y}$, so letting $\mathbf{x}$ and $\mathbf{y}$ run through the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_k$, we get $\mathbf{e}_i \cdot A\mathbf{e}_j = \mathbf{e}_i \cdot A'\mathbf{e}_j$. That says A and A' have the same (i, j) -component for all i and j , so $A = A'$. $\square$

Example 1.2. Taking $k = 2$, $3x^2 - xy + 5y^2 = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} 3 & -1/2 \\ -1/2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

Example 1.3. More generally, $ax^2 + bxy + cy^2 = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$.

Example 1.4. In the ternary case,

$$xy + xz + yz = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

When we write a quadratic form $Q(\mathbf{x})$ as $\mathbf{x} \cdot A\mathbf{x}$ with A being a symmetric matrix, set $B(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot A\mathbf{y}$. This is a bilinear form: $B(\mathbf{x}, \mathbf{y})$ is linear in $\mathbf{x}$ when $\mathbf{y}$ is fixed and it is linear in $\mathbf{y}$ when $\mathbf{x}$ is fixed.² It is also symmetric in $\mathbf{x}$ and $\mathbf{y}$ since $\mathbf{x} \cdot A\mathbf{y} = A^\top \mathbf{x} \cdot \mathbf{y} = A\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot A\mathbf{x} = B(\mathbf{y}, \mathbf{x})$.

Example 1.5. If $Q(\mathbf{x}) = x_1^2 + x_2^2 + \dots + x_k^2 = \mathbf{x} \cdot \mathbf{x}$, then $B(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$.

Example 1.6. If $Q(x, x') = 5x^2 - 14xx' + 10x'^2 = \begin{pmatrix} x \\ x' \end{pmatrix} \cdot \begin{pmatrix} 5 & -7 \\ -7 & 10 \end{pmatrix} \begin{pmatrix} x \\ x' \end{pmatrix}$, then $B((x, x'), (y, y')) = \begin{pmatrix} x \\ x' \end{pmatrix} \cdot \begin{pmatrix} 5 & -7 \\ -7 & 10 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix} = 5xy - 7xy' - 7x'y + 10x'y'$.

The polynomials $Q(\mathbf{x})$ and $B(\mathbf{x}, \mathbf{y})$ determine each other: $Q(\mathbf{x}) = B(\mathbf{x}, \mathbf{x})$ and

$$\begin{aligned} Q(\mathbf{x} + \mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) &= (\mathbf{x} + \mathbf{y}) \cdot A(\mathbf{x} + \mathbf{y}) - \mathbf{x} \cdot A\mathbf{x} - \mathbf{y} \cdot A\mathbf{y} \\ &= \mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} \\ &= B(\mathbf{x}, \mathbf{y}) + B(\mathbf{y}, \mathbf{x}) \\ &= 2B(\mathbf{x}, \mathbf{y}), \end{aligned}$$

²The term “form” in quadratic form and bilinear form means a scalar-valued function.

so $B(\mathbf{x}, \mathbf{y}) = \frac{1}{2}(Q(\mathbf{x} + \mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}))$. This is a higher-dimensional generalization of the identity $xy = \frac{1}{2}((x + y)^2 - x^2 - y^2)$.

Exercises.

1. Show $M\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot M^\top \mathbf{y}$ for all $k \times k$ matrices M and vectors $\mathbf{x}$ and $\mathbf{y}$ with k components.
2. Check $ax^2 + bxy + cy^2 = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} a & b \\ b/2 & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$. Comparing that with Example 1.3, the matrices $\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}$ and $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ are not equal when $b \neq 0$. Why doesn't this contradict the uniqueness of A in Theorem 1.1?
3. Show Theorem 1.1 has a counterexample over every field K characteristic 2: there is *no* 2×2 symmetric matrix A with entries in K such that $x^2 + xy + y^2 = \mathbf{x} \cdot A\mathbf{x}$.

2. DEFINITENESS

The sum of squares quadratic form $x_1^2 + x_2^2 + \cdots + x_k^2$ only has positive values on $\mathbf{R}^k$ except at $\mathbf{0} = (0, \dots, 0)$, where it is 0. Here is a generalization of this example.

Definition 2.1. A k -dimensional quadratic form $Q(\mathbf{x})$ over $\mathbf{R}$ is called *positive-definite* if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbf{R}^k$ and $Q(\mathbf{x}) = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Example 2.2. Set $Q(x, y) = x^2 - xy + y^2$. By completing the square,

$$x^2 - xy + y^2 = \left(x - \frac{y}{2}\right)^2 + 3\frac{y^2}{4} = x'^2 + 3y'^2$$

where $x' = x - y/2$ and $y' = y/2$. Since $x'^2 + 3y'^2 \geq 0$ for all x' and y' and $x'^2 + 3y'^2 = 0$ only when $x' = 0$ and $y' = 0$, which is equivalent to $x = 0$ and $y = 0$,

Example 2.3. Let $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_k^2} = \sqrt{\mathbf{x} \cdot \mathbf{x}}$ be the Euclidean norm on $\mathbf{R}^k$. For an invertible real $k \times k$ matrix M , set $Q(\mathbf{x}) = \|M\mathbf{x}\|^2 = M\mathbf{x} \cdot M\mathbf{x}$. The vector $M\mathbf{x}$ has components that are linear combinations of the components of $\mathbf{x}$, so $M\mathbf{x} \cdot M\mathbf{x}$ is a homogeneous degree 2 polynomial in the components of $\mathbf{x}$: it is a quadratic form. From its definition, $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, and if $Q(\mathbf{x}) = 0$ then $\|M\mathbf{x}\| = 0$, so $M\mathbf{x} = \mathbf{0}$ and thus $\mathbf{x} = \mathbf{0}$ since M is invertible. This shows $Q(\mathbf{x})$ is positive-definite.

The next theorem says that the previous example of a construction of positive-definite quadratic forms explains *all* positive-definite quadratic forms. The key tool in the proof is the spectral theorem for real symmetric matrices.

Theorem 2.4. Let $Q(\mathbf{x})$ be a k -dimensional quadratic form over $\mathbf{R}$. If Q is positive-definite then there is an $k \times k$ invertible real symmetric matrix M such that $Q(\mathbf{x}) = \|M\mathbf{x}\|^2 = M\mathbf{x} \cdot M\mathbf{x}$ for all $\mathbf{x} \in \mathbf{R}^k$.

Proof. Since $Q(\mathbf{x})$ is a quadratic form over $\mathbf{R}$, by Theorem 1.1 we can write $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ for a symmetric $k \times k$ real matrix A , and A is invertible: if A were *not* invertible there would be a nonzero $\mathbf{x} \in \mathbf{R}^k$ such that $A\mathbf{x} = \mathbf{0}$. Then $Q(\mathbf{x}) = \mathbf{x} \cdot \mathbf{0} = 0$, which by positive-definiteness implies $\mathbf{x} = \mathbf{0}$: contradiction.

The spectral theorem for real symmetric matrices says that every such matrix has a basis of orthogonal eigenvectors. Thus A has a basis $\mathbf{v}_1, \dots, \mathbf{v}_k$ where $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ and $i \neq j$ then $\mathbf{v}_i \perp \mathbf{v}_j$, so $\mathbf{v}_i \cdot \mathbf{v}_j = 0$. We can scale the $\mathbf{v}_i$'s so they all have length 1. Then $\mathbf{v}_1, \dots, \mathbf{v}_k$ is

an orthonormal basis of $\mathbf{R}^k$, not just an orthogonal basis: they are mutually perpendicular with length 1, not just mutually perpendicular.

From the formula for $Q(\mathbf{x})$ as the dot product $\mathbf{x} \cdot A\mathbf{x}$, $Q(\mathbf{v}_i) = \mathbf{v}_i \cdot A\mathbf{v}_i = \lambda_i(\mathbf{v}_i \cdot \mathbf{v}_i)$ for all i . The dot product $\mathbf{v}_i \cdot \mathbf{v}_i$ is $\|\mathbf{v}_i\|^2 = 1$ and $Q(\mathbf{v}_i) > 0$ since Q is positive-definite. Then $\lambda_i = Q(\mathbf{v}_i)/(\mathbf{v}_i \cdot \mathbf{v}_i) = Q(\mathbf{v}_i)$, so all the eigenvalues of A are positive when $\mathbf{x} \cdot A\mathbf{x}$ is positive-definite.

Let C be the change of basis matrix from the standard basis $\mathbf{e}_1, \dots, \mathbf{e}_k$ of $\mathbf{R}^k$ to the eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_k$ of A : $C\mathbf{e}_i = \mathbf{v}_i$ for all i , so the columns of C are $\mathbf{v}_1, \dots, \mathbf{v}_k$. Since the columns of C are an orthonormal basis of $\mathbf{R}^k$, we have dot product formulas $\mathbf{v}_i \cdot \mathbf{v}_j = 1$ if $i = j$ and 0 if $i \neq j$. That implies $C^\top C = I_k$, so $C^{-1} = C^\top$.

Let $D = \text{diag}(\lambda_1, \dots, \lambda_k)$ be the diagonal matrix whose i th diagonal entry is the eigenvalue of A for $\mathbf{v}_i$: $D\mathbf{e}_i = \lambda_i\mathbf{e}_i$. Then $AC = CD$ since the two sides here are linear on $\mathbf{R}^k$ and agree on the basis $\mathbf{e}_1, \dots, \mathbf{e}_k$: the common value of AC and CD at each $\mathbf{e}_i$ is $\lambda_i\mathbf{v}_i$ (check!). From $AC = CD$, we have $A = CDC^{-1}$. The eigenvalues of A are all positive, so write $\lambda_i = s_i^2$ where $s_i > 0$. Then $D = S^2$ where $S = \text{diag}(s_1, \dots, s_k)$, so $A = CDC^{-1} = CS^2C^{-1} = (CSC^{-1})^2 = (CSC^\top)^2$ and $CSC^\top$ is symmetric: $(CSC^\top)^\top = CS^\top C^\top = CSC^\top$ since S , being diagonal, is its own transpose.

Set $M = CSC^{-1} = CSC^\top$, so $M = M^\top$ and $A = M^2$, so A being invertible implies M is invertible. This implies

$$Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x} = \mathbf{x} \cdot M^2\mathbf{x} = M^\top \mathbf{x} \cdot M\mathbf{x} = M\mathbf{x} \cdot M\mathbf{x} = \|M\mathbf{x}\|^2,$$

which is what we wanted to show. $\square$

Corollary 2.5. *When $Q(\mathbf{x})$ is a k -dimensional positive-definite quadratic form over $\mathbf{R}$, its level sets $\{\mathbf{x} : Q(\mathbf{x}) = c\}$ for each $c \geq 0$ are bounded. In particular, the equation $Q(\mathbf{x}) = c$ has finitely many solutions in $\mathbf{Z}^k$.*

Proof. By Theorem 2.4, there is a $k \times k$ invertible real (symmetric) matrix M such that $Q(\mathbf{x}) = \|M\mathbf{x}\|^2$ for all $\mathbf{x}$ in $\mathbf{R}^k$. Then $Q(\mathbf{x}) = c$ is the same as $\|M\mathbf{x}\| = \sqrt{c}$. The set $S := \{\mathbf{v} : \|\mathbf{v}\| = \sqrt{c}\}$ is a sphere in $\mathbf{R}^k$ of radius $\sqrt{c}$ and

$$Q(\mathbf{x}) = c \iff \mathbf{x} \in M^{-1}(S_{\sqrt{c}}).$$

The sphere $S_{\sqrt{c}}$ is compact and since $M^{-1} : \mathbf{R}^k \rightarrow \mathbf{R}^k$ is linear it is continuous (Exercise 2.3), so the image $M^{-1}(S_{\sqrt{c}})$ in $\mathbf{R}^k$ is compact. Thus this image is bounded. $\square$

A quadratic form over $\mathbf{R}$ that is not positive definite (and not negative definite, which is the same as positive definite except positivity of values away from the origin are replaced by negativity) can have unbounded level sets, so these may contain infinitely many integral points. Consider $Q(x, y) = x^2 - 2y^2$ and the 1-level set $\{(x, y) \in \mathbf{R}^2 : x^2 - 2y^2 = 1\}$, which has infinitely many integral points since Pell's equation has infinitely many integral solutions.

Definition 2.6. For a k -dimensional integral positive-definite quadratic form Q , its *representation numbers* are $r_Q(n) := |\{\mathbf{x} \in \mathbf{Z}^k : Q(\mathbf{x}) = n\}|$ for integers $n \geq 0$.

When $Q(x_1, \dots, x_k) = x_1^2 + x_2^2 + \dots + x_k^2$, the numbers $r_Q(n)$ are written as $r_k(n)$: this is the number of k -tuples of integers whose squares add up to n .

Example 2.7. Let $Q(x, y) = x^2 + y^2$ and $r_2(n) = \{\mathbf{x} \in \mathbf{Z}^2 : Q(\mathbf{x}) = n\}$. The table below gives values of $r_2(n)$ for $0 \leq n \leq 12$. The integral solutions to $x^2 + y^2 = 10$ are $(\pm 1, \pm 3)$ and $(\pm 3, \pm 1)$ with independent choices of sign in each coordinate.

n	0	1	2	3	4	5	6	7	8	9	10	11	12
$r_2(n)$	1	4	4	0	4	8	0	0	4	4	8	0	0

An important way to study the representation numbers $r_Q(n)$ as n varies is through analytic properties of a suitable generating function for this sequence. Here are two such generating functions.

Zeta-function of Q : for complex s , set

$$\zeta_Q(s) = \sum_{\mathbf{x} \in \mathbf{Z}^k, \mathbf{x} \neq \mathbf{0}} \frac{1}{Q(\mathbf{x})^s} = \sum_{n \geq 1} \frac{r_Q(n)}{n^s}.$$

This series turns out to be absolutely convergent when $\operatorname{Re}(s) > k/2$. It has a meromorphic continuation to $\mathbf{C}$ and a functional equation resembling that of the Riemann zeta-function. But unlike the Riemann zeta-function, $\zeta_Q(s)$ usually does *not* have an Euler product or satisfy a Riemann hypothesis.

Theta-function of Q : for τ in the upper half-plane $\mathfrak{h} = \{z = x + iy : x, y \in \mathbf{R}, y > 0\}$, set

$$\theta_Q(\tau) = \sum_{\mathbf{x} \in \mathbf{Z}^k} e^{\pi i Q(\mathbf{x})\tau} = \sum_{n \geq 0} r_Q(n) e^{\pi i n \tau}.$$

(Writing $\tau = x + iy$, $|e^{\pi i n \tau}| = |e^{\pi i n x} e^{-\pi n y}| = e^{-\pi n y}$ gets small when $y > 0$ is big.) This is convergent on all of $\mathfrak{h}$, where it is analytic. It turns out to be a modular form for a suitable subgroup of $\operatorname{SL}_2(\mathbf{Z})$.

Both $\zeta_Q(s)$ and $\theta_Q(\tau)$ depend on Q just through the numbers $r_Q(n)$ as n varies. When we make an invertible *integral* linear change of variables³ (like $(x, y) \mapsto (x + y, y)$, with inverse $(x, y) \mapsto (x - y, y)$, but not $(x, y) \mapsto (x + y, x - y)$ with inverse $(x, y) \mapsto ((x + y)/2, (x - y)/2)$, where the inverse map isn't defined over $\mathbf{Z}$), a quadratic form Q turns into another quadratic form Q' with the same representation numbers, so $\zeta_Q = \zeta_{Q'}$ and $\theta_Q = \theta_{Q'}$. For example, $(x, y) \mapsto (x + y, y)$ turns $Q(x, y) = x^2 + y^2$ into $Q'(x, y) = (x + y)^2 + y^2 = x^2 + 2xy + 2y^2$, so $r_Q(n) = r_{Q'}(n)$ for all integers $n \geq 0$. Is a converse true? That is, if Q and Q' are two positive-definite integral quadratic forms of the same dimension and $r_Q(n) = r_{Q'}(n)$ for all $n \geq 0$, are Q and Q' linked by an invertible integral linear change of variables? The answer in general is *no*: in 1964, Milnor gave the first counterexamples in dimension 16. In 1992, Conway and Sloane found a counterexample in dimension 4: the quadratic forms

$$Q(x, y, z, t) = 2x^2 + 4y^2 + 6z^2 + 8t^2 - xy - 4xz - 4xt - 4yt + zt$$

and

$$Q'(x, y, z, t) = 2x^2 + 4y^2 + 6z^2 + 8t^2 + 2xy + 2xz + 5xt - 3yz + 2yt$$

have the same zeta-function and the same theta-function, but they are not linked to each other by an invertible integral linear change of variables.

Exercises.

1. Let $f(x_1, \dots, x_k) \in \mathbf{R}[\mathbf{x}_1, \dots, \mathbf{x}_k]$ be homogeneous in k variables with *even* degree. Show that if $f(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbf{Z}^k$ then $f(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbf{R}^k$; the converse is obvious. In particular, a k -dimensional quadratic form over $\mathbf{Z}$ that is nonnegative on $\mathbf{Z}^k$ is nonnegative on $\mathbf{R}^k$. (Homogeneity of f is important here: $6x^2 - 6x + 1$ is not homogeneous and is positive on all of $\mathbf{Z}$ but it is not positive on all of $\mathbf{R}$.)

³A linear change of variables is described by a matrix, and saying the change of variables is invertible over $\mathbf{Z}$ means the matrix is integral with determinant ± 1 .

2. Show a binary quadratic form $ax^2 + bxy + cy^2$ is positive-definite if and only if $a > 0$ and $b^2 - 4ac < 0$, or equivalently $c > 0$ and $b^2 - 4ac < 0$. (Hint: complete the square.)
3. When $L : \mathbf{R}^k \rightarrow \mathbf{R}^k$ is a linear transformation, show L is continuous. (Hint: look at the formulas for the component functions of L .) Your argument should be extendable to the case of linear transformations $\mathbf{R}^k \rightarrow \mathbf{R}^\ell$ where k and ℓ might be different: such functions are continuous. Can you show linear transformations between Euclidean spaces are uniformly continuous?
4. Find an invertible *integral* linear change of variables that converts $5x^2 - 14xy + 10y^2$ into $x^2 + y^2$.